The Atom

Exam code: 0625 & 0972

Download PDF

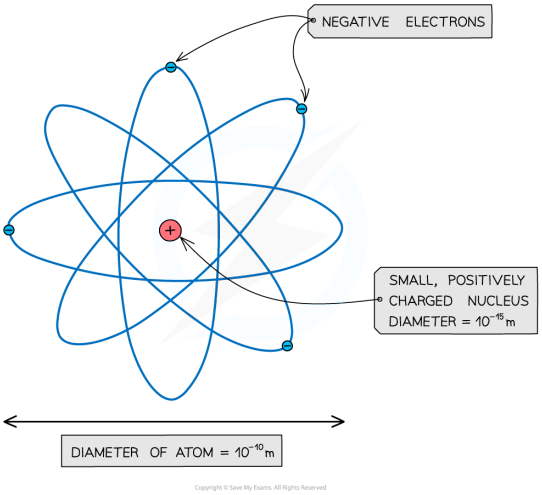
Written by: Ashika
Reviewed by: Caroline Carroll
Updated on 10 February 2026

Did this video help you? Yes No

Atomic structure

- Atoms are the building blocks of **all matter**
- They consist of a small dense **positively** charged nucleus and **negatively** charged electrons in orbit around the nucleus

Atomic structure



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

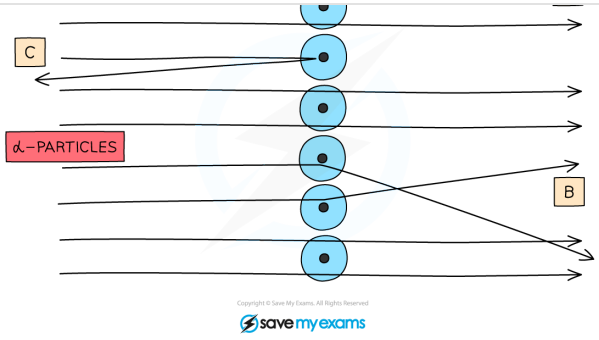
- Atoms are incredibly small, with a radius of only 1×10^{-10} m
  - This means that about one hundred million atoms could fit side by side across your thumbnail
- The radius of the nucleus is over 10,000 times smaller than the whole atom, but it contains almost **all of the mass** of the atom

Rutherford's alpha scattering experiment

Extended tier only

- In Rutherford's alpha scattering experiment the scattering of alpha (α) particles by a sheet of thin metal supports the **nuclear model** of the atom, by providing evidence for:
  - a very **small nucleus** surrounded by **mostly empty space**
  - a **nucleus** containing most of the **mass** of the atom
  - a nucleus that is **positively charged**
- In 1909 a group of scientists were investigating the Plum Pudding model of atomic structure
  - Physicist, **Ernest Rutherford** was instructing two of his students, Hans Geiger and Ernest Marsden to carry out the alpha scattering experiment
- They directed a beam of **alpha particles** (He^{2+} ions) at a thin gold foil
  - They expected the alpha particles to **travel through** the gold foil, and maybe change direction a small amount
- As shown in the diagram, instead they discovered that :
  - Most of the alpha particles **passed straight through** the foil because the atom is mostly **empty space** (A)
  - Some of the alpha particles were **deflected** (changed direction**)** as they passed through the foil because they came close to the positively charged nucleus, which repelled the positively charged alpha particles (B)
  - A few of the alpha particles **bounced back** off the gold foil because the nucleus is tiny
- This bouncing back could not be explained by the Plum Pudding model of atomic structure, so a new model had to be created
  - This was the first evidence of the nuclear atom as atomic structure

Rutherford's alpha scattering experiment results

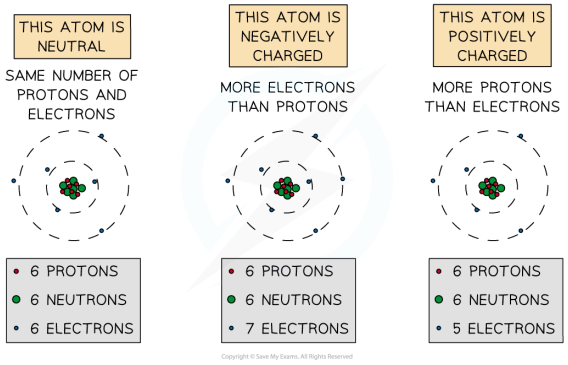


When α-particles are fired at thin gold foil, most of them go straight through but a very small number bounce straight back

Atoms & ions

- An ion is an **electrically charged** atom or group of atoms formed by the **loss** or **gain** of **electrons**
  - An atom will lose or gain electrons to become more stable
- A stable atom is normally electrically neutral
  - This means it has the same number of protons (positive charge) and electrons (negative charge)
- Positive ions are therefore formed when atoms **lose** electrons
  - There will be more protons than electrons
- Negative ions are therefore formed when atoms **gain** electrons
  - There will be more electrons than protons

The atomic structure of ions



The difference between positive and negative ions

overall charge of the atom. If an atom has 5 protons, 5 neutrons and 6 electrons, it has a **net** negative charge because it's a negative ion (more electrons than protons).

Remember which way around the charges are by **p**roton being **p**ositive.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself Flashcards

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity

Subjects

Biology
Chemistry

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.